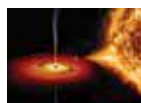




Behold the wondrous sights

A compilation of some of the strangest and the most awe-inspiring images of Mars taken to date. <https://digit.in/jun20-03>



Black hole emission

NASA's Chandra Observatory observed a black hole emitting hot material into space. <https://digit.in/jun20-04>

or the farthest distance from Jupiter. The tidal forces at that point would squeeze plumes from deep subsurface oceans, a rare but breathtaking sight. Anyone on the surface would see a gushing mist of ice particles going straight up to the skies, as the sporadic plumes reach altitudes in excess of 200 kilometers, more than 20 times the height of mount Everest. The geysers are also ten times more energetic than their counterparts on the Saturnian moon of Enceladus, throwing out 2000 kg of material every second. The plumes rise from cracks in the ice crust, in a landscape of criss-crossing striations known as lineae. The ground is so tortured, that it is known as the chaos terrain. If any explorer were to fire up an app that includes a temperature reading in the picture, then it would read a cool -200°C . The temperature is so cold that tiny particles of water in the atmosphere would be a mist of razor sharp ice crystals, something of a hazard for any humans around, even when protected by spacesuits.

Even at the maximum distance, Jupiter dominates the sky, tilted at an angle with even the features visible, including the Giant Red Spot and the aurorae circling the north pole. If someone were to camp at the location for a few days, they would see the moons of Amalthea, Metis and Thebe making circuits around the Gas Giant, with Io trailing far behind and Gany-mede making the occasional appearance. If the conditions are just right, then Earth, Venus, and Mercury would line up in the night sky, but would probably be difficult to spot because of the glare from the Sun.

Landing the spacecraft on the surface would be another tricky operation in itself, because of the 50 foot tall spikes on the surface. These spikes, called pentatites are pointed towards the Sun and sharpened by the action of the heat. The water sublimates away without turning into a gaseous form, so even the water vapour is made of ice. What remains behind are essentially towering blades of ice that can easily decimate fast-moving spacecraft

approaching too close to the surface. This is no place for a ski resort.

A COMET

Visiting a comet would be an interesting prospect for a number of reasons. They are not the snowballs that people imagined them to be for so long, as the Rosetta spacecraft showed us with 67P/Churyumov-Gerasimenko, which has been described as a dumb-bell or a duck. The cometary cores can be all kinds of strange shapes. Scientists believe that planets may in fact tend to shift towards having two lobes just like 67P/Churyumov-Gerasimenko because of the low gravity combined with the rotation.

cause of heat from the Sun. The stars would be blocked out because of what would essentially look like a fog.

MORDOR

Who would not want to disobey Boromir and walk into Mordor? The New Horizons team informally referred to a large dark spot on Charon, the largest moon of Pluto as Mordor. It is not an official name yet, but the name is likely to stick because it is already in widespread use. It is believed that material from Pluto gets deposited near the north pole of the moon. The dark shade is believed to be caused by compounds known as Tholins, which are organic in nature and considered to be the build-



Geysers on Europa

For explorers on the surface, the low gravity would cause all kinds of weird effects. Because of the odd shape, the gravitational pull would not always be directed downwards. People might find themselves falling sideways, or perhaps even stumbling upwards between lobes, but the traditional directions would not even make too much sense here. The falling would actually be very, very slow. According to Space.com, an object dropped from chest height would take a full two minutes to touch the surface. Visitors would have to be very careful while moving, as it is possible to simply escape the gravitational influence of the comet by jumping too high.

Oh, then there is the nebula, the cloud of particles constantly being stripped away from the surface be-

ing blocks of life. Do not worry though, you will not be bumping into any aliens, fell beasts or orcs, as the other ingredients necessary for life have not been identified on Charon, at least, not yet. The moon appears to have active geology as well as a differentiated core. The light from the Sun would still light up the deep red landscape, but at that extreme distance, it would just appear just a little larger than the other stars in the sky. The Milky Way would be clearly visible at all times though, even during the day. If you have some time to spare, you might as well make a visit to the informally Serenity Chasma because you took the trouble of going all the way to Charon. It is a 200 kilometer long, 40 kilometer wide and 5 kilometer deep fault line, and yes, it is named after the ship from the Firefly TV show. 